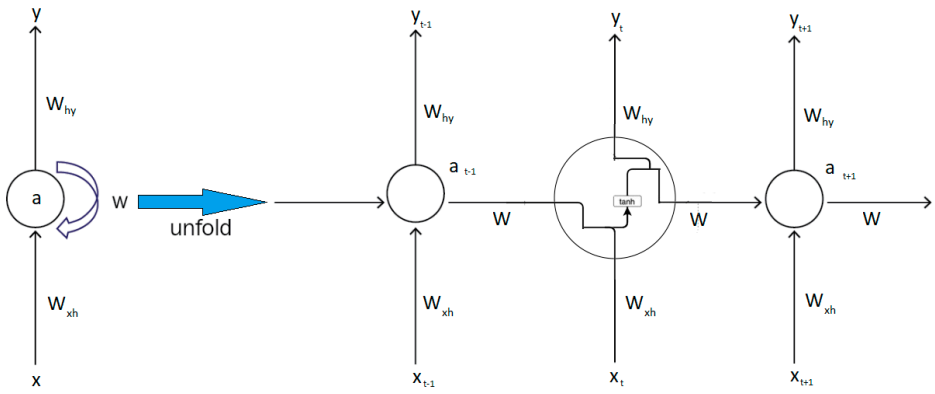
TEL354 - Minería de datos

Clase 17: Large Language Models

Prof. Nicolás Torres
nicolas.torresr@usm.cl
Ingeniería Civil Telemática
Departamento de Electrónica
Universidad Técnica Federico Santa María

Neurona recurrente

$$h_t = f(W_{xh}x_t + W_{hh}h_{t-1} + b_h); \qquad y_t = W_{hy}h_t + b_y$$

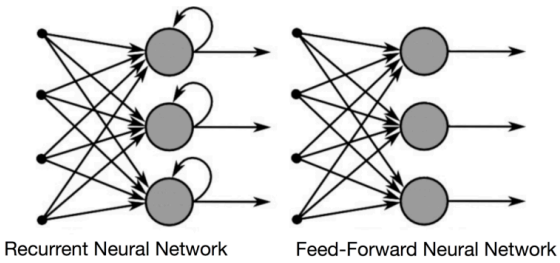


Capas recurrentes

- SimpleRNN layer
- LSTM layer
- GRU layer
- Bidirectional layer
- etc.

Capa Recurrente

Una capa **Recurrente** (`SimpleRNN`) es una Fully-connected RNN donde la salida en tiempo t, se usa para alimentar la entrada en tiempo $t+1$.



El problema de las dependencias a largo plazo

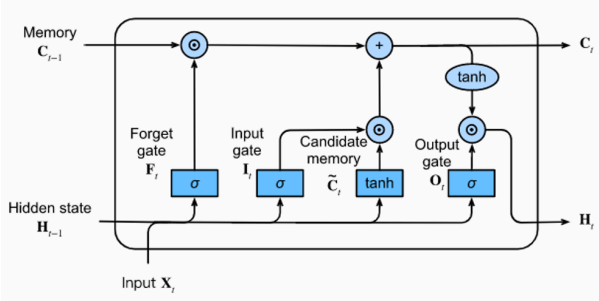
Las RNNs sufren de memoria a corto plazo y van olvidando las primeras entradas de la secuencia.

```
... and that's why now I speak fluent _____.
```

Para mitigar este problema, se crearon dos redes neuronales recurrentes especializadas: Long Short-Term Memory (LSTM) y Gated Recurrent Units (GRU). Ambas son capaces de aprender secuencias a largo plazo, utilizando compuertas lógicas que realizan diferentes operaciones tensoriales, permitiendo aprender qué información mantener y qué información eliminar en la capa oculta.

Long Short-Term Memory

En una LSTM, cada neurona mantiene un estado de memoria de la celda c_t en el tiempo t y tiene tres compuertas: de olvido, de entrada y de salida.



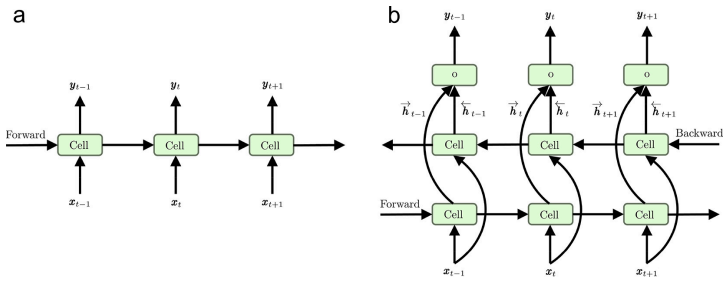
El problema del contexto pasado y futuro

```
Apple is my favorite _____ ... (fruit?/company?/phone?)

Apple is my favourite _____, and I work there.
Apple is my favorite _____, and I am going to buy one.
```

Bidirectional recurrent neural networks (BRNN)

Procesan secuencias hacia adelante y hacia atrás (o de izquierda a derecha y viceversa).



Las Bi-RNN (b) consisten en dos RNN (a) separadas que procesan los datos de entrada en direcciones opuestas, y las salidas de estas RNN se combinan para producir la salida final.

In [1]:
```
# pip install transformers
```

In [2]:
```
from transformers import pipeline, logging

logging.set_verbosity_error()
```

In [3]:
```
BERT = pipeline('fill-mask', model='bert-base-uncased')
```

In [4]:
```
BERT('A golden retriever is a [MASK].')
```

Out[4]:
```
[{'score': 0.7484612464904785,
  'token': 3899,
  'token_str': 'dog',
  'sequence': 'a golden retriever is a dog.'},
 {'score': 0.044763702899217606,
  'token': 13314,
  'token_str': 'mascot',
  'sequence': 'a golden retriever is a mascot.'},
 {'score': 0.023076245561242104,
  'token': 4743,
  'token_str': 'bird',
  'sequence': 'a golden retriever is a bird.'},
 {'score': 0.02024541236460209,
  'token': 4937,
  'token_str': 'cat',
  'sequence': 'a golden retriever is a cat.'},
 {'score': 0.016355333849787712,
  'token': 8843,
  'token_str': 'breed',
  'sequence': 'a golden retriever is a breed.'}]
```

Limitación BERT

BERT muestra un gran rendimiento con su naturaleza bidireccional; sin embargo, parece quedarse corto en la comprensión de la negación.

In [5]:
```
BERT('A golden retriever is not a [MASK].')
```

Out[5]:
```
[{'score': 0.7157090306282043,
  'token': 3899,
  'token_str': 'dog',
  'sequence': 'a golden retriever is not a dog.'},
 {'score': 0.039168745279312134,
  'token': 13314,
  'token_str': 'mascot',
  'sequence': 'a golden retriever is not a mascot.'},
 {'score': 0.016858208924531937,
  'token': 8843,
  'token_str': 'breed',
  'sequence': 'a golden retriever is not a breed.'},
 {'score': 0.01580878533422947,
  'token': 4937,
  'token_str': 'cat',
  'sequence': 'a golden retriever is not a cat.'},
 {'score': 0.013033190742135048,
  'token': 4743,
  'token_str': 'bird',
  'sequence': 'a golden retriever is not a bird.'}]
```

Variaciones de BERT

Existen muchas variaciones del modelo BERT. Un Lite BERT (ALBERT), por ejemplo, es una versión liviana de BERT donde la cantidad de parámetros (x18 menos parámetros) y el tiempo de entrenamiento (x1.7 más rápido) se reducen. El enfoque BERT robustamente optimizado (RoBERTa) es otra variación en la que las mejoras se realizan esencialmente entrenando BERT en un conjunto de datos más grande con lotes más grandes. Existen otras variaciones para diferentes idiomas, como BETO en español.

In [6]:
```
fill_mask = pipeline("fill-mask",
                     model="dccuchile/bert-base-spanish-wwm-cased",
                     tokenizer="dccuchile/bert-base-spanish-wwm-cased")
```

In [7]:
```
fill_mask("El español es un idioma muy [MASK] en el mundo.")
```

Out[7]:
```
[{'score': 0.38821306824684143,
  'token': 7856,
  'token_str': 'hablado',
  'sequence': 'El español es un idioma muy hablado en el mundo.'},
 {'score': 0.11342769861221313,
  'token': 7022,
  'token_str': 'utilizado',
  'sequence': 'El español es un idioma muy utilizado en el mundo.'},
 {'score': 0.08567869663238525,
  'token': 21712,
  'token_str': 'extendido',
  'sequence': 'El español es un idioma muy extendido en el mundo.'},
 {'score': 0.07988113164901733,
  'token': 6362,
  'token_str': 'popular',
  'sequence': 'El español es un idioma muy popular en el mundo.'},
 {'score': 0.054361846297979355,
  'token': 9234,
  'token_str': 'usado',
  'sequence': 'El español es un idioma muy usado en el mundo.'}]
```

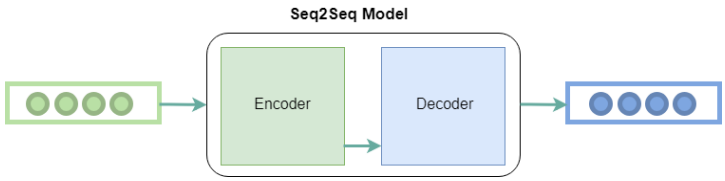
In [8]:
```
fill_mask("Lavarse frecuentemente las manos con agua y [MASK].")
```

Out[8]:
```
[{'score': 0.9864119291305542,
  'token': 24556,
  'token_str': 'jabón',
  'sequence': 'Lavarse frecuentemente las manos con agua y jabón.'},
 {'score': 0.005320871714502573,
  'token': 1435,
  'token_str': 'sal',
  'sequence': 'Lavarse frecuentemente las manos con agua y sal.'},
 {'score': 0.0013185136485844 85,
  'token': 9536,
  'token_str': 'aceite',
  'sequence': 'Lavarse frecuentemente las manos con agua y aceite.'},
 {'score': 0.0009088400402106345,
  'token': 7585,
  'token_str': 'azúcar',
  'sequence': 'Lavarse frecuentemente las manos con agua y azúcar.'},
 {'score': 0.000565165770240128,
  'token': 26639,
  'token_str': 'limón',
  'sequence': 'Lavarse frecuentemente las manos con agua y limón.'}]
```

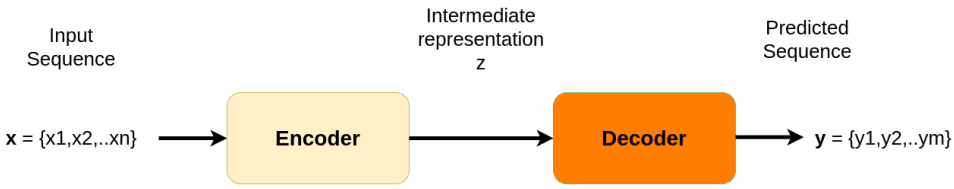
In [9]:
```
fill_mask("Fui a la librería a comprar un [MASK].")
```

Out[9]:
```
[{'score': 0.6376195549964905,
  'token': 3872,
  'token_str': 'libro',
  'sequence': 'Fui a la librería a comprar un libro.'},
 {'score': 0.030473770573735237,
  'token': 7153,
  'token_str': 'periódico',
  'sequence': 'Fui a la librería a comprar un periódico.'},
 {'score': 0.02276957966387272,
  'token': 7265,
  'token_str': 'regalo',
  'sequence': 'Fui a la librería a comprar un regalo.'},
 {'score': 0.020322028547525406,
  'token': 23621,
  'token_str': 'boleto',
  'sequence': 'Fui a la librería a comprar un boleto.'},
 {'score': 0.019596733152866364,
  'token': 1958,
  'token_str': 'poco',
  'sequence': 'Fui a la librería a comprar un poco.'}]
```

Sequence to Sequence (Seq2Seq) Model



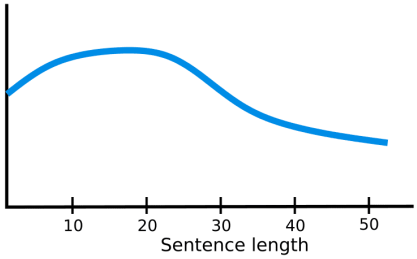
Sequence to sequence learning



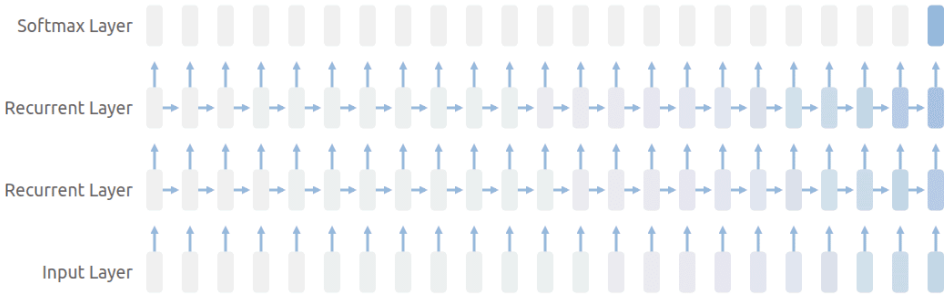
Problema con seq2seq learning

La representación intermedia **z** no puede codificar información de todos los instantes de tiempo en la secuencia de entrada. Esto se conoce comúnmente como el problema del cuello de botella.

En teoría, las matemáticas indican que es posible capturar toda la información sobre la secuencia de entrada. Sin embargo, en la práctica, hasta dónde podemos ver en el pasado (la llamada ventana de referencia) es finita. Por lo tanto, las RNN tienden a olvidar información de los instantes de tiempo que están muy atrás.



Problema del desvanecimiento de gradiente (Vanishing gradient problem)



Vanishing Gradient: where the contribution from the earlier steps becomes insignificant in the gradient for the vanilla RNN unit.

Atención al rescate!

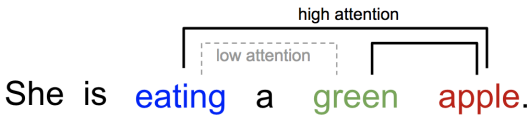


La idea central es que el vector de contexto $\mathbf{z}$ debería tener acceso a todas las partes de la secuencia de entrada en lugar de solo a la última. En otras palabras, necesitamos formar una conexión directa con cada instante de tiempo.

Esto es lo que ahora llamamos atención, que es simplemente una noción de memoria, obtenida al prestar atención a múltiples entradas a lo largo del tiempo.

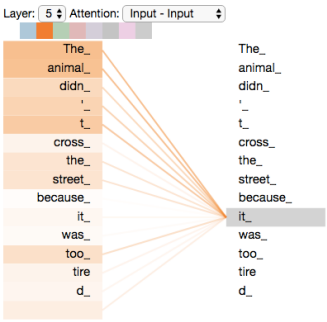
Atención explícita

La red pondera su sensibilidad a la entrada en función de la memoria de entradas anteriores.

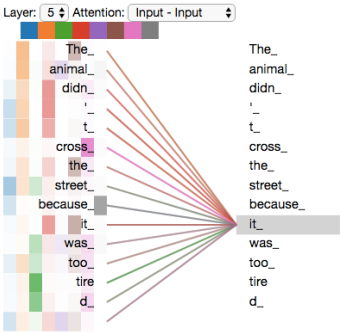


Self-attention

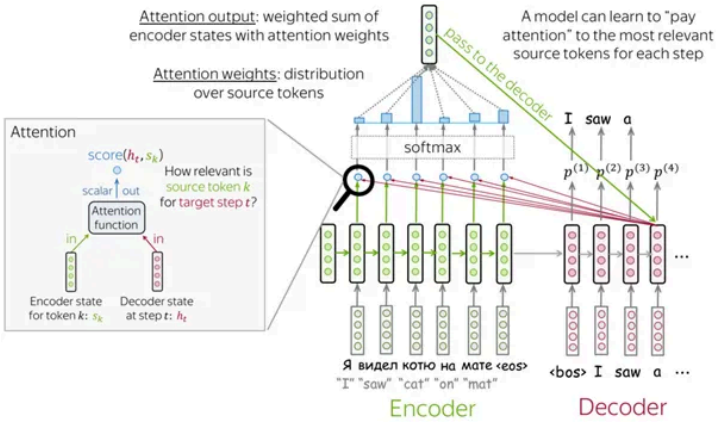
También podemos definir la atención de la misma secuencia, llamada autoatención. En lugar de buscar una asociación/alineación de secuencia de entrada-salida, ahora buscamos puntuaciones entre los elementos de la secuencia, como se muestra a continuación:



Multi Head Attention



Mecanismo de atención



In [10]:
```
GPT2 = pipeline('text-generation', model='gpt2')
```

In [11]:
```
gen_text = GPT2('A golden retriever is a', num_return_sequences=5,
                max_length=100, pad_token_id=GPT2.tokenizer.eos_token_id
```

In [12]:
```
for k in gen_text:
    print(k['generated_text'].replace('\n',''),"\n")
```

```
A golden retriever is a rare and valuable animal. However, its true strength
is not limited to a handful of special traits. As a human, this dog has devel
oped great patience, quick judgment, and persistence. When learning to walk u
pright or sitting upright, it usually knows when to try a new pose or to lie
down so it can stay on its back.A golden retriever, in addition to its abilit
y to walk, is also an athlete's companion. Just ask Chris Paul,

A golden retriever is a beloved member of the California Golden Retriever Con
servation Fund [PDF], but it appears to be doing some serious damage too. Sin
```

ce the Golden Retriever (RPRRJ) is an endangered vertebrate, its numbers have increased by 75% in the past two years and its population has been increasing by about 40%.The California National Guard's Wildlife Resources Management Branch is now investigating possible violations of a permit by the California Department of Fish and Game to harvest

A golden retriever is a good example of a breed that could make an ideal puppy. If you want to learn more about the different breeds of golden retrievers, then look up the Golden Retriever Puppy's Guide, at http://toyshode.com/guide/golden_grunt.The Golden Retriever will have two eyes, which are about the same width as the average blue retriever's. Each eyespan has a different length, and one

A golden retriever is a very unusual breed. It can be the last of its kind for much of history, having been bred from around 1755 when, as the English term was originally used for it they weren't given any kind of care nor was it ever bred indoors or out. It can live up to 150 years and can usually be found at least in part-time for a year, which makes it a very nice summer residence.Golden retrievers are an interesting breed, both

A golden retriever is a magnificent animal, but her habitat seems much smaller — just a few hundred cubs.She doesn't get an ounce of carefree attention.Instead, she gets her attention from her owners and their dogs. She can't go for a long walk in the backyard anymore, because she couldn't hold on hard enough, so she gets her attention by using her paws.The puppy, however, gets the most attention — she gets to look like

In [13]:
```
gen_text = GPT2('A golden retriever is not a', num_return_sequences=5,
                    max_length=100, pad_token_id=GPT2.tokenizer.eos_token_id
```

In [14]:
```
for k in gen_text:
    print(k['generated_text'].replace('\n',''),"\n")
```

A golden retriever is not a regular part of a zoo or zoo rescue. It is rather common, however, that the Golden Retriever and other golden retriever species are in the wrong enclosure when in a small room, where others may find it difficult to handle or carry the animals in case of rescue.Golden Retriever are known to breed at higher rate while in captivity, and in captivity, in most parts of the United States, their numbers generally fall by as much as

A golden retriever is not a domesticated wild horse, although, for breeding purposes, it might be considered domesticated. It is also considered an uneman cipated wild horse, or "wild horse". It does not include the "white" race (Hexelian horse) which is domesticated.Hexelia has a common ancestor within the Great Lakes and has been domesticated for over 4,000 years, most certainly as a result of domestication by several

A golden retriever is not a "dog" but a "human," as it "is" or as it is affectionately put. It actually has no dog's name or nameplate. So a gold retriever is defined by its color and markings like white. A black or gray-gray is defined by its neck shape. In other words, gold retriever may be defined by its coloring and some may be defined as "green." While there are exceptions (such as Labrador Retrievers

A golden retriever is not a regular breed, but if you're looking for some sweet treats hypnotic (he's a real treat), I suggest that you take in some of these things instead. Just in case, if you're looking for something more serious, the most important thing to remember is that the only part of your body that you have control over is the head and neck (except the eyes when the dogs are inside).Here's some details on why you need a leash to

A golden retriever is not a normal bird. It lacks the right eyes or ears. It is usually called a "fellow dog".Although a golden retriever is known to be very cute, many people take notice. The "tongue in the hay" sign that has been used many times to greet Golden Retriever is called a jowl. It is sometimes r

eferred to as "gatorade".But the jowl, which is not quite as big as